LECTURE 4:
LEBESGUE INTEGRATION

The goal of these lecture notes is to give an exposition of the material in the same order and approximately in the same completeness as it was done in class (ORF526). Always refer to the textbooks if in doubt and for more complete treatment.

Please email me with any observed typos to `elre@princeton.edu`, thanks in advance!

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Notation: Given an underlying measure space $(\Omega, \mathcal{F}, \mu)$, let $\mathcal{SF}_+$ denote the set of all non-negative simple functions:

$$\mathcal{SF}_+ := \left\{ \sum_{n=1}^{N} c_n \mathbb{1}_{A_n}(\omega), \quad \text{for all } N \geq 1, c_n \geq 0, A_n \in \mathcal{F} \text{ for } n = 1, \dots N \right\}.$$

Definition 4.1. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. Let $f : \Omega \to \bar{\mathbb{R}}$ be a measurable function (where $\bar{\mathbb{R}} := \mathbb{R} \cup \{\pm\infty\}$ denotes the extended reals). The **Lebesgue integral** of f with respect to the measure μ, denoted by $\int_\Omega f(\omega)\, \mathrm{d}\mu(\omega)$ or $\int_\Omega f(\omega)\, \mu(\mathrm{d}\omega)$, is defined via the following steps:

- Step 1. For **non-negative simple functions**: If $f(\omega) = \sum_{n=1}^{N} c_n \mathbb{1}_{A_n}(\omega) \in \mathcal{SF}_+$, then
$$\int_\Omega f(\omega)\ \mathrm{d}\mu(\omega) := \sum_{i=1}^{N} c_i \mu(A_i).$$
- Step 2. For **non-negative functions**: If $f : \Omega \to [0, \infty]$, then
$$\int_\Omega f(\omega) d\mu(\omega) := \sup_{g \in \mathcal{SF}_+,\, 0 \leq g \leq f} \left\{ \int_\Omega g(\omega) d\mu(\omega) \right\}.$$
- Step 3. For **arbitrary measurable functions**: Define the positive and negative parts of f by $f_+(\omega) := \max\{f(\omega), 0\}$ and $f_-(\omega) := -\min\{f(\omega), 0\}$ respectively. Both f_+ and f_- are non-negative functions and so we can define
$$\int_\Omega f(\omega)\ \mathrm{d}\mu(\omega) := \int_\Omega f_+(\omega)\ \mathrm{d}\mu(\omega) - \int_\Omega f_-(\omega)\ \mathrm{d}\mu(\omega).$$

Clearly, the last difference is well-defined only if at least one of the two integrals of f^+ or f^- is finite. Hence, we say that a measurable function f is **integrable** only if **both** integrals of f_+ and f_- are finite. Note that f is (Lebesgue) integrable if and only if $|f|$ is (Lebesgue) integrable.

We will usually omit ω and simply write $\int f\ \mathrm{d}\mu = \int_\Omega f(\omega)\ \mathrm{d}\mu(\omega)$ (just like with the probability that a random variable takes a certain value). We will also just say "integrable" to refer to Lebesgue integrable functions.

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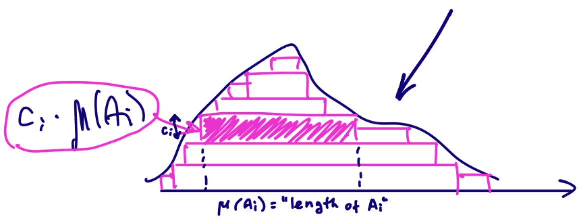
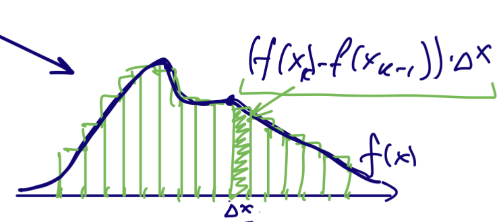
Remark 4.2.
- For Step 2, recall that any non-negative measurable function is a limit of non-negative simple functions (from the monotone approximation by simple functions theorem in Lecture 3).
- To check that the definition of a Lebesgue integral is well-posed in Step 1, we also need to check that two different representations of a simple function as a finite sum of indicators result in the same value of the integral. This can be shown along the following lines: if $\sum_{i=1}^{N} c_i \mathbb{1}_{A_i} = \sum_{j=1}^{M} d_j \mathbb{1}_{B_j}$, then there exists at most 2^{N+M} (finitely many) disjoint sets C_ℓ so that each A_i and B_j is a union of some C_ℓ. The summation over the same or disjoint indicators must have coinciding values for pointwise equality.

Lebesgue and Riemann integrals





- (When do Riemann and Lebesgue integrals coincide?) If f is a bounded function on a compact set and Riemann-integrable, then it is Lebesgue-integrable and the integrals match (we won't prove this).
- (A Lebesgue integrable function that is not Riemann integrable) Consider the Dirichlet function $f : [0,1] \to \{0,1\}$ such that $f(x) = 1$ if and only if x is rational. It is not Riemann integrable but $\int f \, \mathrm{d}x = 0$ with respect to the Lebesgue measure (e.g., this follows from property (1) of Proposition 4.3 below, noting that the rational numbers form a countable subset with Lebesgue measure zero).
- *Notation:* If we do not specify, we will always consider Lebesgue integrals with respect to the Lebesgue measure. However, note that more generally, the Lebesgue integral has the flexibility to be defined with respect to any measure. For example, one can consider the *Dirac measure* δ_a corresponding to some $a \in \mathbb{R}$, such that $\delta_a(A) = 1$ if and only if $a \in A$ (check that this is a well-defined measure). Then for any measurable f, we have
$$\int_{\mathbb{R}} f \, \mathrm{d}\delta_a(x) = f(a).$$

Properties of Lebesgue integrals

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. If a property holds for all inputs ω except possibly on a subset of Ω of measure zero, we say that it holds *almost everywhere* and write *a.e.* to denote it. When we talk about random variables and probability spaces, we frequently say that an event happens *almost surely* (or *a.s.*) instead if it fails with probability zero.

Proposition 4.3. *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, and f, g be integrable functions from Ω to $\bar{\mathbb{R}}$. The following properties hold:*

(1) ***Nice behavior almost everywhere.***
- *If f is non-negative, then $\int_\Omega f \,\mathrm{d}\mu = 0$ if and only if $f = 0$ almost everywhere.*
- *$f = g$ a.e. (that is, $\mu(\{\omega : f(\omega) \neq g(\omega)\}) = 0$) implies $\int f \,\mathrm{d}\mu = \int g \,\mathrm{d}\mu$.*
- *$|f| < \infty$ a.e.*

(2) ***Monotonicity.*** *If $f \leq g$ (almost) everywhere, then $\int f \,\mathrm{d}\mu \leq \int g \,\mathrm{d}\mu$.*

(3) ***Linearity.*** *For any $\alpha, \beta \in \mathbb{R}$, the function $\alpha f + \beta g$ is also integrable, and*

$$\int (\alpha f + \beta g) \,\mathrm{d}\mu = \alpha \int f \,\mathrm{d}\mu + \beta \int g \,\mathrm{d}\mu.$$

(4) ***Absolute continuity.*** *For any $\varepsilon > 0$, there exists $\delta > 0$ so that for any integrable function f and any measurable set A with $\mu(A) < \delta$, the integral $\int_A |f| \,\mathrm{d}\mu < \varepsilon$.*

Proof of (1). We will only verify the first part here, and leave the other two parts as exercises. Which direction of part one fails if we do not assume non-negativity of f?

Let f be a non-negative integrable function. First, suppose that $\mu(\{\omega : f(\omega) > 0\}) = 0$. For any non-negative simple function $g \leq f$, note that

$$\{\omega : g(\omega) > 0\} \subseteq \{\omega : f(\omega) > 0\},$$

so both these sets have measure zero. For a simple function, a direct check shows that this implies that $\int g \,\mathrm{d}\mu = 0$. Since this holds for any such simple function, this also holds for their supremum, and therefore we conclude that $\int f \,\mathrm{d}\mu = 0$.

Conversely, suppose that $\mu(\{\omega : f(\omega) > 0\}) > 0$. Consider

$$A_n := \{\omega : f(\omega) > n^{-1}\}.$$

Note that by continuity from below,

$$\mu(\{\omega : f(\omega) > 0\}) = \mu\left(\bigcup_{n=1}^{\infty} (\{\omega : f(\omega) > n^{-1}\}\right) = \lim_{n\to\infty} \mu(A_n).$$

Hence, $\mu(A_n) > 0$ for some n, and so

$$\int f \,\mathrm{d}\mu \geq \int f \mathbb{1}_{A_n} \,\mathrm{d}\mu \geq \int n^{-1} \mathbb{1}_{A_n} \,\mathrm{d}\mu = n^{-1}\mu(A_n) > 0.$$

We used (everywhere) monotonicity in the first inequality. □

Proof sketch of (2). Note that since $g \leq f$, the set of non-negative simple functions lower-bounding f is a subset of the set of non-negative functions lower-bounding g. So, the first supremum is less than or equal to than the second.

If f can be negative, we can write $f = f_+ - f_-$ and the same argument can be applied to its positive f_+ and negative parts f_- respectively.

If the inequality holds *almost* everywhere, then we need to involve part (1). □

Proof sketch of (3). The proof of linearity proceeds via a ***standard 4-step argument*** (sometimes called the *"standard machine"*), which consists of steps similar to the ones used in the definition of the Lebesgue integral:

(1) A property is checked for *indicator functions.*
(2) By linearity, the property is extended to all *non-negative simple functions.*

(3) The property is extended to all *non-negative measurable functions* by writing it as an increasing limit of non-negative simple functions (by the monotone approximation theorem from Lecture 3), and using the monotone convergence theorem to interchange the limit and integral (this is stated precisely in the next section).
(4) Finally, the property is extended to all *integrable functions* by writing $f = f_+ - f_-$ and using linearity. □

The property (4) *can be proved* using the dominated convergence theorem below (Theorem 4.5)). This is a smooth version of an informal statement that the sets of measure zero do not matter for the integration.

Convergence theorems for exchanging limits and integrals

Theorem 4.4 (Monotone convergence theorem (MCT)). *Suppose that*

$$f(\omega) = \lim_n f_n(\omega) \quad \textit{and} \quad f_1(\omega) \le f_2(\omega) \le \dots \quad \textit{for all } \omega;$$

that is, $\{f_n\}_n$ *is a pointwise increasing sequence of functions* $\Omega \to [0, \infty]$ *that converges pointwise to* f*. Then* f *is measurable and*

$$\int f \, \mathrm{d}\mu = \lim_n \int f_n \, \mathrm{d}\mu.$$

Proof. First, convince yourself (easy from what we already know!) that f must be measurable. To prove the equality, we will prove two inequalities. One side is easy: from monotonicity,

$$\int f \, \mathrm{d}\mu \ge \int f_n \, \mathrm{d}\mu.$$

Thus,

$$\int f \, \mathrm{d}\mu \ge \lim_n \int f_n \, \mathrm{d}\mu.$$

For the reverse inequality:

- Check (exercise!) that for any simple nonnegative function $g = \sum_{n=1}^N c_n \mathbb{1}_{B_n}$ on the measure space $(\Omega, \mathcal{F}, \mu)$, the set function

$$\nu_g(A) := \int_\Omega g \mathbb{1}_A \, \mathrm{d}\mu = \sum_{n=1}^N c_n \mathbb{1}_{B_n} \mathbb{1}_A$$

  is a measure. We denote the same integral as $\nu_g(A) = \int_A g \, \mathrm{d}\mu$. (Actually, this holds for any measurable function g, but this extension requires the monotone convergence theorem that we are now proving.)
- Consider any simple function g such that $0 \le g \le f$. For any $\alpha \in (0, 1)$, consider

$$S_n(\alpha) := \{\omega : \alpha g(\omega) \le f_n(\omega)\}.$$

  Note that for any $\alpha < 1$,

$$\bigcup_{n=1}^\infty S_n(\alpha) = \Omega \quad \text{and} \quad S_1(\alpha) \subseteq S_2(\alpha) \subseteq \dots$$

(Note that for $\alpha = 1$ we might have that all of the sets are empty and do not converge to Ω.) Then, by continuity of the measure ν_g,

$$\nu_g(S_n(\alpha)) \to \nu_g(\Omega).$$

- So, for any simple function g such that $0 \leq g \leq f$, we have

$$\begin{aligned}\int_\Omega g \, \mathrm{d}\mu = \nu_g(\Omega) = \lim_{n\to\infty} \nu_g(S_n(\alpha)) &= \lim_{n\to\infty} \int_{S_n(\alpha)} g \, \mathrm{d}\mu \\ &\leq \lim_{n\to\infty} \int_{S_n(\alpha)} \frac{f_n}{\alpha} \, \mathrm{d}\mu \leq \frac{1}{\alpha} \lim_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mu.\end{aligned}$$

  This holds for any $\alpha \in (0,1)$, so by taking $\alpha \to 1$, we have

$$\int_\Omega g \, \mathrm{d}\mu \leq \lim_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mu.$$

- Finally, by the definition of the Lebesgue integral,

$$\begin{aligned}\int_\Omega f \, \mathrm{d}\mu &= \sup_{g\in\mathcal{SF}_+ :\, 0\leq g\leq f} \left\{ \int_\Omega g \, \mathrm{d}\mu \right\} \\ &\leq \sup_{g\in\mathcal{SF}_+ :\, 0\leq g\leq f} \left\{ \lim_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mu \right\} = \lim_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mu. \qquad \square\end{aligned}$$

If we are given a sequence of (integrable) functions that is not necessarily pointwise increasing and non-negative, we can also exchange the limit and the integral if it is uniformly dominated by an integrable function:

Theorem 4.5 (**Dominated convergence theorem (DCT)**). *Suppose that*

$$f(\omega) = \lim_n f_n(\omega) \quad \textit{and} \quad |f_n(\omega)| \leq g(\omega) \ \textit{for all } \omega, \ \textit{where } g \textit{ is integrable};$$

that is, $\{f_n\}_n$ is a sequence of integrable functions $\Omega \to \bar{\mathbb{R}}$ that converges pointwise to f, which is uniformly pointwise upper-bounded (dominated) by another integrable function g. Then f is also integrable and

$$\int f \, \mathrm{d}\mu = \lim_n \int f_n \, \mathrm{d}\mu.$$

Moreover, one can show the stronger result $\lim_n \int |f - f_n| \, \mathrm{d}\mu = 0$.

The dominated convergence theorem can be derived using the following result (which is also important on its own).

Theorem 4.6 (**Fatou's lemma**). *Suppose that $\{f_n\}_n$ is a sequence of non-negative measurable functions $\Omega \to [0,\infty]$. Then*

$$\int (\liminf_{n\to\infty} f_n) \, \mathrm{d}\mu \leq \liminf_{n\to\infty} \int f_n \, \mathrm{d}\mu.$$

In turn, Fatou's lemma can be proved using the monotone convergence theorem.

The inequality can be strict, ege, consider a sequence of functions $f(x) = n$ for $[0, n^{-1}]$ and 0 otherwise, on $[0,1]$.

Remark 4.7.
- Note that the non-negativity assumption often substitutes for integrability (e.g., in the monotone convergence theorem, and also in the properties of the Lebesgue integral). The integral of a non-negative measurable function can always be defined (however, it may be infinite).
- The monotone convergence theorem and the dominated convergence theorem are also true if we only assume that the hypotheses are valid *almost everywhere*.

Integration over a subset and probability density functions

Given an non-negative integrable function $f : \mathbb{R} \to [0, \infty]$, we define the Lebesgue integral of f over a measurable set A by

$$\int_A f \, \mathrm{d}\mu := \int_\Omega f \mathbb{1}_A \, \mathrm{d}\mu = \int_\Omega f(\omega)\mathbb{1}_A(\omega) d\mu(\omega).$$

This is how the probability density function of a random variable was defined in Lecture 3: if the law $\mathcal{P}_X$ of a random variable X on $(\Omega, \mathcal{F}, \mathbb{P})$ can be represented by

$$\mathcal{P}_X(A) := \mathbb{P}(X \in A) = \int_A f_X \, \mathrm{d}\lambda$$

for some function f_X and with λ being the Lebesgue measure on $\mathbb{R}$, then we say that $f_X : \mathbb{R} \to \mathbb{R}$ is the probability density function of X.

Note that:

(a) *Every such integral always defines a measure.* Formally, this should be be checked using "the standard machine" used in the proof of Proposition 4.3: first check that countable additivity holds for set functions defined via integrals of non-negative simple functions $f \in \mathcal{SF}_+$, then use the monotone convergence theorem to extend the result to integrable functions.

So, given a non-negative function $f \geq 0$ (pointwise) such that $\int_{\mathbb{R}} f \, \mathrm{d}\lambda = 1$, the integral above defines a probability measure on $\mathbb{R}$, $\nu_f := \int_A f \, \mathrm{d}\lambda$. This measure can be taken to be the law of some random variable. For example, one can define $(\Omega, \mathcal{F}, \mathbb{P}) := (\mathbb{R}, \mathcal{B}(\mathbb{R}), \nu_f)$ and set X to be identity map $\mathbb{R} \to \mathbb{R}$, which defines a random variable X with the given density f.

(b) *Not every measure coming from the law of a random variable can be represented this way* (i.e., not all random variables have densities). For example, if a probability measure has atoms (sets of Lebesgue measure zero with non-zero probability), which is true, e.g., for any discrete distribution, then its law cannot be represented as an integral with respect to the Lebesgue measure, and so it does not have density. One way to see it is to show, using Proposition 4.3, that an integral over a set of measure zero (like an one point set with respect to a Lebesgue uniform measure) must be zero. However, note that we might be able to still write such a measure as an integral with respect to some other measure (e.g., the Dirac or counting measure).